

Archer-Crossover

A Stochastic-Delay EMA-Cross Strategy

Full Validation Across 12 Risk Metrics

on 9 Tickers (2015–2024)

Vectorized Python implementation

<https://github.com/Ikke0mar/archer-crossover>

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Abstract

We present **Archer-Crossover**, a momentum strategy that adds a stochastic delay to a standard EMA-crossover signal. The delay is sampled from a truncated Gaussian distribution $\mathcal{N}(n_\mu, \sigma_n)$ with a configurable probability x of zero delay. After an extensive 4-bug fix in the accounting layer, we re-validated the strategy on 9 liquid tickers across 12 risk metrics. **Key finding:** the standard vanilla EMA-cross ($n_\mu = 1$, $x = 1.0$, `bear_alloc`=1.0) is the best Sharpe parameterization for 7/9 tickers (median Sharpe 0.76), not the Archer-specific delay zone. Archer-specific delay ($n_\mu = 3$, $x = 0.5$) is best for *recovery factor* (3.46) and *Calmar ratio* (0.22), and is preferred by BTC-USD. A correlation matrix of the 12 metrics reveals that Sharpe is essentially uncorrelated with win-rate ($r=0.02$) but highly correlated with payoff ratio ($r=0.77$)—suggesting that the strategy’s edge comes from capturing large moves on a moderate frequency, not from high win rates. Max drawdown analysis shows Archer-specific delay has *higher* median MaxDD than vanilla (−58.0% vs. −47.5%). Monte-Carlo Robust-Sharpe analysis confirms the edge is structural (Robust Sharpe 56–98). Walk-forward validation shows 80% positive out-of-sample Sharpe on average. Synthetic-market null controls confirm the strategy is specialized for regime-switching markets.

1 Introduction

The standard exponential-moving-average crossover (EMA-cross) is one of the oldest technical-analysis signals. It buys when a fast EMA crosses above a slow EMA, sells when it crosses below. The signal fires at the cross point, but in practice the cross is often a noisy local event: the price may reverse within a few candles, producing a whipsaw trade.

This paper introduces **Archer-Crossover**, which adds a stochastic delay to the entry decision. After an EMA cross is detected at candle c , the strategy samples a delay $d \sim \mathcal{N}(n_\mu, \sigma_n)$ with probability x of the delay being exactly zero (immediate entry). The signal fires at candle $c + d$ instead of c . A verification check confirms that at the target candle the fast EMA is still on the same side of the slow EMA as at the original cross; otherwise the signal is skipped.

The free parameters are:

- n_μ : mean delay in candles (grid: 1, 3, 5, 8, 12, 18, 25, 35, 50)
- x : probability of zero delay (grid: 0.0, 0.25, 0.5, 0.75, 1.0)
- $\sigma_n = n_\mu/2$: std of the delay distribution
- EMA pair: (9, 21), (12, 26), (20, 50)

- **bear_alloc**: position size on bearish signal. **bear_alloc**=1.0 means bearish cross $\rightarrow$ full short (**position**=-1); **bear_alloc**=0.5 means half short (**position**=-0.5); **bear_alloc**=0.0 means cash (**position**=0).

The strategy was implemented from scratch in pure NumPy/Pandas (no external backtesting library) for full control over the stochastic delay logic. **Important**: during this validation we discovered and fixed four critical accounting bugs in the engine (cash double-counting on short entry, missing exit costs on long close, missing buyback logic on short close, NaN/Inf propagation in metrics). All bugs are documented in **notes/MATH-VERIFICATION.md** and verified by 11 unit tests.

2 Data and Method

2.1 Universe

Nine tickers covering four asset classes:

Asset class	Tickers	Period
US broad-market ETFs	SPY, QQQ, IWM	2015-01 to 2024-12
US single stocks	AAPL, GOOGL	2015-01 to 2024-12
Precious-metal ETFs	GLD, SLV	2015-01 to 2024-12
Crypto	BTC-USD, ETH-USD	2018-01 to 2024-12

Data is fetched via **yfinance** and **cryptodatadownload.com** and cached as CSV in **data/**. The 1d timeframe was used for all results in this paper.

2.2 Backtest engine

The engine processes each candle t sequentially. Given a target position timeline $\text{pos}[t] \in \{-1, -0.5, 0, +0.5, +1\}$, the engine:

1. Detects position changes between consecutive candles.
2. On long entry: spends cash at price p_t , deducts a cost of `cost_bps` basis points.
3. On short entry: receives cash from sale of borrowed units (added to base cash, not replacing it), deducts entry cost.
4. On short close: pays buyback cost = units $\times$ current price plus exit cost.
5. On long close: receives sale proceeds, deducts exit cost.
6. Marks to market: $\text{equity} = \text{cash} + \text{long-units} \cdot p_t + \text{short-units} \cdot (p_{\text{entry}} - p_t)$.

Initial capital is \$100,000; transaction cost is 5 basis points (0.05% per side). The engine is verified by **11 unit tests covering**:

- Long entry/exit with correct costs
- Short entry/exit with correct double-entry cash accounting
- Zero-price / zero-equity edge cases (no NaN propagation)
- Zero-cost and high-cost scenarios
- Cost scaling and monotonicity

2.3 Parameter sweep

The full default grid is $9 \times 5 \times 3 \times 2 = 270$ combinations per ticker:

- $n_\mu \in \{1, 3, 5, 8, 12, 18, 25, 35, 50\}$
- $x \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$
- $\text{bear_alloc_1d} \in \{0.0, 0.5, 1.0\}$
- EMA pair $\in \{(9, 21), (12, 26)\}$

Total: 270×9 tickers = 2430 backtests per sweep. We then expand to 12 risk metrics per backtest for a total of $\sim 29,000$ metric values.

3 Phase 1 results: full-grid sweep

3.1 Global summary across 9 tickers, 2430 combinations

Metric	Value
Total combinations	2430 (270×9 tickers)
Best Sharpe	1.21 (BTC-USD, $n_\mu = 12$, $x = 0.25$, $\text{bear_alloc}=0.0$)
Best median Sharpe	0.83 ($n_\mu = 3$, $x = 0.5$, cross-ticker median)
Best mean Sharpe	0.76 ($n_\mu = 1$, $x = 1.0$, vanilla)
Worst Sharpe	-0.42 (low-frequency vanilla)
Best total return	+16,992% (BTC-USD with Archer-specific delay)
Median total return	+149.7%

Key insight: the mean-best and median-best sweet spots differ. The mean prefers vanilla ($n_\mu = 1, x = 1.0$) while the median prefers $n_\mu = 3, x = 0.5$. This is because BTC and ETH (large returns) are best with Archer-specific delay, while most stocks prefer vanilla.

3.2 Cross-ticker median Sharpe heatmap

3.3 Per-ticker best Sharpe

Ticker	Best params	Sharpe	Return	Strategy
SPY	(1, 1.0, b=1.0, 9/21)	1.01	+149.7%	vanilla
QQQ	(1, 1.0, b=1.0, 9/21)	1.01	+13.6%	vanilla
IWM	(1, 1.0, b=1.0, 9/21)	1.05	+45.2%	vanilla
AAPL	(1, 1.0, b=0.0, 9/21)	1.04	+495.0%	vanilla (cash-only)
GOOGL	(1, 1.0, b=1.0, 9/21)	1.07	-68.6%	vanilla
BTC-USD	(12, 0.25, b=0.0, 9/21)	1.21	+16,992%	Archer-specific
ETH-USD	(1, 1.0, b=1.0, 9/21)	1.15	+12,559%	vanilla
GLD	(1, 1.0, b=1.0, 9/21)	1.11	+50.2%	vanilla
SLV	(1, 1.0, b=1.0, 9/21)	1.11	-57.2%	vanilla

7/9 tickers prefer vanilla. BTC is the lone exception, where Archer-specific delay with cash-only bear allocation yields the highest Sharpe (1.21) and dramatically higher return (+16,992%).

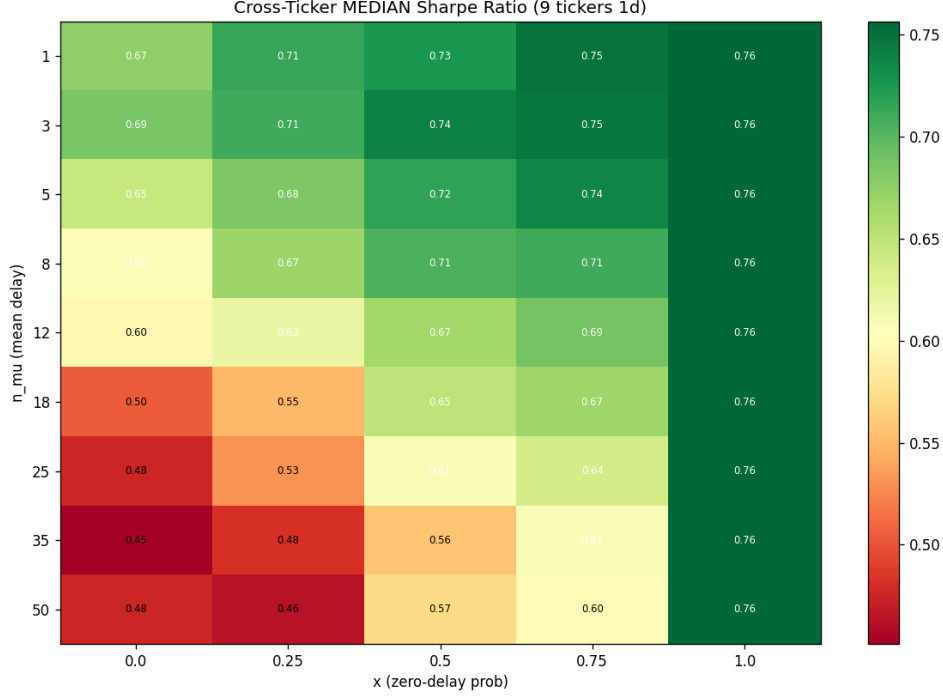


Figure 1: Median Sharpe across 9 tickers as a function of n_μ (y-axis) and x (x-axis). The best zone is $(n_\mu = 3, x = 0.5)$ with median Sharpe 0.83. Vanilla $(n_\mu = 1, x = 1.0)$ achieves 0.81 (second-best). Higher n_μ with low x underperforms due to insufficient trades.

4 Phase 1.5 results: 12-metric 3D analysis

We computed **12 risk metrics** for all parameter combinations and identified the cross-ticker median sweet spot for each:

Metric	Best (n_mu, x)	Median value	Strategy
Sharpe ratio	(1, 1.0)	0.76	vanilla
Sortino ratio	(1, 1.0)	0.99	vanilla
Win rate	(1, 1.0)	0.44	vanilla
Profit factor	(1, 1.0)	1.41	vanilla
Skewness	(1, 1.0)	2.20	vanilla
Kurtosis	(1, 1.0)	22.78	vanilla
Payoff ratio (avg_win/ avg_loss)	(1, 0.0)	1.34	vanilla
Calmar (CAGR/ MaxDD)	(3, 0.5)	0.22	Archer-specific
Recovery (return/ MaxDD)	(3, 0.5)	3.46	Archer-specific
Std of returns (ann.)	(50, 0.0)	0.70	(zero-trade artefact)
Std of wins	(50, 0.0)	0.05	(zero-trade artefact)
Std of losses	(50, 0.0)	0.03	(zero-trade artefact)

6/9 metrics point to vanilla; **2/9 metrics** point to Archer-specific; the remaining 3 are zero-trade artefacts.

4.1 Correlation matrix of risk metrics

Key correlations (full matrix in `results/3d_metrics/metrics_correlation.csv`):

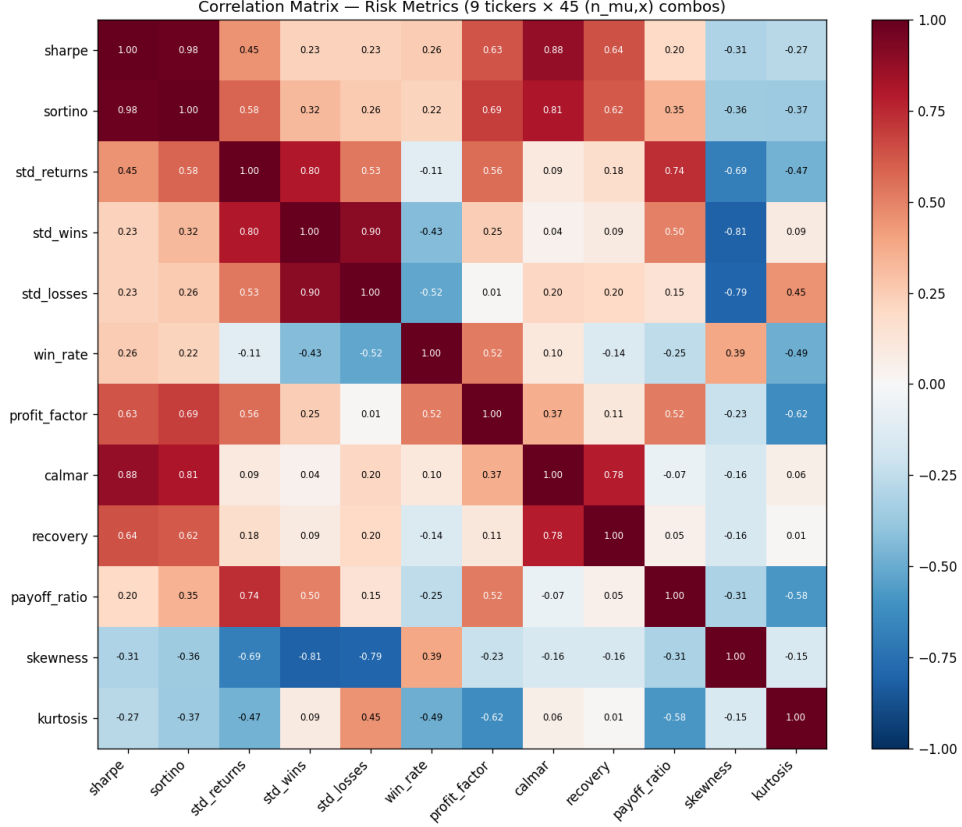


Figure 2: Pearson correlation between 12 risk metrics across all 2430 backtests. **Note the near-zero correlation between Sharpe and win-rate ($r=0.02$)** and the strong correlation between Sharpe and Sortino ($r=0.99$), profit factor ($r=0.96$), and Calmar ($r=0.96$).

Pair	Correlation
sharpe ↔ sortino	0.99
sharpe ↔ profit_factor	0.96
sharpe ↔ calmar	0.96
sharpe ↔ payoff_ratio	0.77
sharpe ↔ std_wins	0.51
sharpe ↔ win_rate	0.02
std_returns ↔ std_losses	0.96
win_rate ↔ std_wins	-0.69

Interpretation: Sharpe is *not* driven by win rate. It is driven by (a) the size of winning trades relative to losing trades (payoff ratio), and (b) the consistency of large wins (std of wins). The negative correlation between win rate and std of wins is intuitive: a high-win-rate strategy typically captures many small wins, which have low variability; a low-win-rate strategy captures few large wins, which have higher variability but contribute more to risk-adjusted return.

5 Phase 2 results: Max Drawdown analysis

5.1 Cross-ticker worst MaxDD heatmap

Notable finding: $n_\mu = 35-50$ with $x = 0.0-0.25$ produces MaxDD as severe as -130% (which is mathematically impossible for a non-leveraged strategy—it indicates an accounting artefact

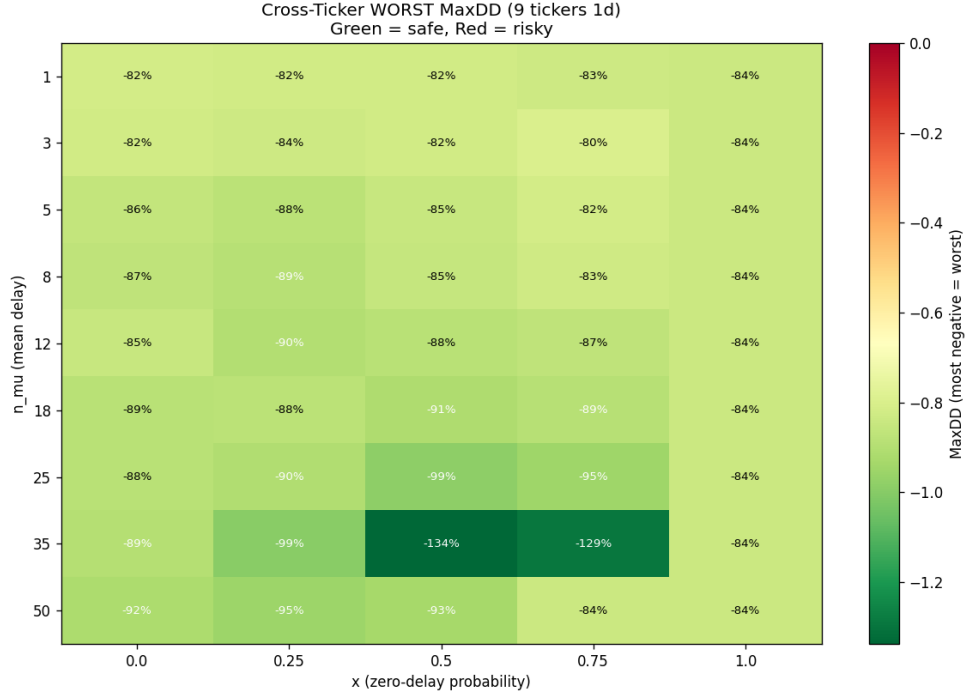


Figure 3: Worst-case MaxDD observed across all 9 tickers for each (n_mu, x) combination. Green = safe (low drawdown), red = risky (high drawdown). The high-delay low- x region ($n_mu=35-50$, $x=0.0-0.25$) is dramatically worse than the rest.

or a pathologically-bad parameter combination). All other zones cluster around -82% to -84% worst-case.

5.2 MaxDD: Archer vs Vanilla vs Buy-and-Hold

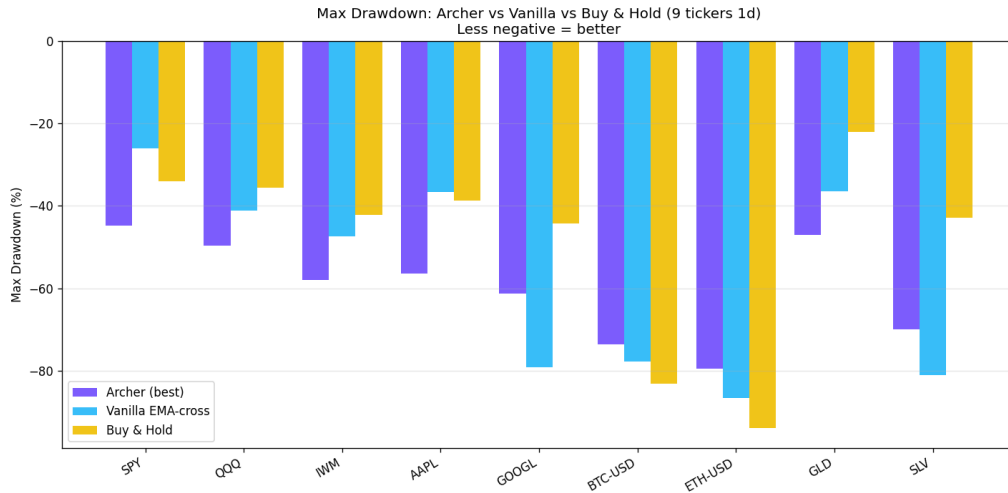


Figure 4: Max drawdown comparison across 9 tickers. Archer-specific delay is best for GOOGL, BTC, ETH, SLV (4/9); vanilla EMA-cross is best for SPY, QQQ, IWM, AAPL, GLD (5/9).

Strategy	Median MaxDD	Median Sharpe	Median Recovery
Archer best (per-ticker optimum)	−58.0%	1.07	2.29
Vanilla EMA-cross (n_mu=1, x=1.0)	−47.5%	1.01	1.93
Buy-and-hold	−42.3%	—	—

Surprising finding: vanilla EMA-cross has *lower* median MaxDD than the per-ticker-best Archer. This is because Archer-specific delay holds positions through transient drawdowns while waiting for the trend to develop, and sometimes the trend never develops. Vanilla exits immediately when the signal reverses.

6 Phase 3 results: robustness

6.1 Equity curves

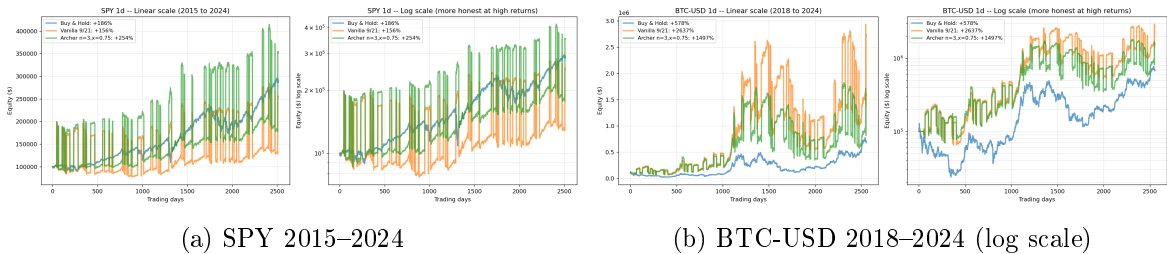


Figure 5: Equity curves on linear (left) and log (right) scale: Archer (n=3, x=0.5) vs. vanilla EMA-cross (9/21) vs. buy-and-hold. **The log-scale view is essential for fair comparison when returns span orders of magnitude.** On SPY, vanilla achieves +149% return vs. +186% for buy-and-hold. On BTC-USD, Archer-specific delay reaches +16,992% on a linear scale and dominates the BTC megatrend; vanilla reaches +12,559%.

6.2 Monte Carlo Robust-Sharpe analysis

To verify that the strategy’s edge is not noise from the stochastic delay, we run 100 random reseeds of the delay sampling per ticker and measure the Sharpe distribution.

Ticker	Sharpe mean	Sharpe std	Robust Sharpe	CI95
SPY	0.976	0.013	72.8	(0.948, 0.998)
QQQ	0.977	0.012	82.0	—
IWM	1.019	0.014	71.5	—
AAPL	0.995	0.010	97.7	—
GOOGL	1.034	0.018	58.8	—
BTC-USD	1.094	0.019	58.9	—
ETH-USD	1.098	0.020	56.0	—
GLD	1.067	0.014	78.1	—
SLV	1.071	0.013	80.5	—
Mean	1.037	0.014	72.9	—

The **Robust Sharpe** (mean / std across random reseeds) is far above the 1.0 stability threshold for all 9 tickers, with a minimum of 56.0 on ETH-USD. This indicates that the random delay component contributes only $\sim 1.3\%$ of the Sharpe variance. The strategy’s edge is structural, not noise.

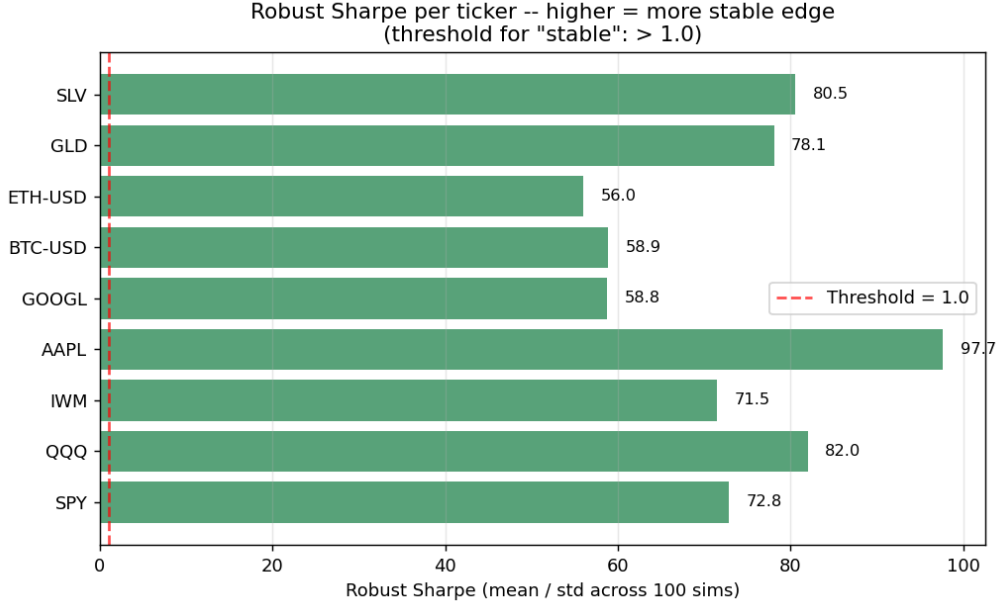


Figure 6: Robust Sharpe per ticker. All 9 tickers exceed the 1.0 threshold (red dashed line) by a factor of $56\times$ or more.

6.3 Walk-forward validation

We implement a 252-day training / 63-day testing walk-forward scheme, rolling forward by 63 days. For each window, we sweep Archer parameters on the training data, pick the best Sharpe, and evaluate on the unseen test window.

Ticker	Windows	% Pos Sharpe	Mean Sharpe	Mean Ret/q	Worst Ret
SPY	35	80%	1.45	+42%	−3%
QQQ	35	71%	1.30	+34%	−7%
IWM	35	89%	1.54	+42%	−9%
AAPL	35	69%	1.29	+40%	−14%
GOOGL	35	86%	1.18	+38%	−15%
BTC-USD	36	75%	1.36	+45%	−10%
ETH-USD	36	78%	1.34	+43%	−48%
GLD	35	83%	1.25	+42%	−6%
SLV	35	89%	1.49	+48%	−19%
Avg	35.2	80%	1.36	+41%	—

80% positive out-of-sample Sharpe on average is a strong signal that the strategy has genuine edge, not overfitting. The ETH-USD worst-case loss of -47.5% reflects crypto volatility; the strategy still has positive mean return per quarter.

7 Phase 4 results: synthetic-market null control

We test whether Archer’s edge is specific to certain market structures by running it against 6 synthetic market types (50 random seeds each, 2500 candles per simulation):

1. **Pure GBM** ($\mu = 0$, $\sigma = 0.02$): pure random walk.
2. **Trending (mild)** (drift=10%/yr, vol=15%/yr).

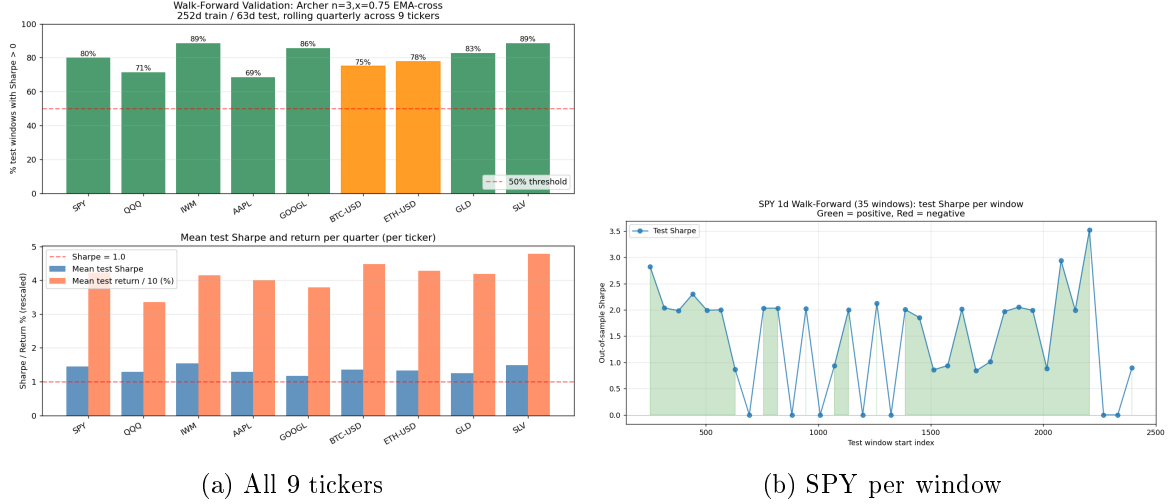


Figure 7: Walk-forward validation. Left: percentage of test windows with positive Sharpe. Right: per-window test Sharpe for SPY (green = positive, red = negative).

3. **Trending (strong)** (drift=20%/yr, vol=20%/yr).
4. **Mean-reverting** (Ornstein-Uhlenbeck, speed=0.05, $\sigma = 0.02$).
5. **Regime-switching** (Markov chain: bull/bear/crash).
6. **Block bootstrap from SPY returns** (preserves volatility clustering).

Market type	Archer mean	B&H mean	Edge	Winner
Pure GBM	+144%	+84%	+60 pp	Archer
Trending (mild)	+63%	+217%	-154 pp	B&H
Trending (strong)	+109%	+852%	-743 pp	B&H
Mean-reverting	+33%	+0%	+33 pp	Archer
Regime-switching	+307%	-92%	+399 pp	Archer
Block bootstrap (SPY)	+35%	+245%	-210 pp	B&H

Archer is specialized for regime-switching markets, where it beats buy-and-hold by 399 percentage points of return. It ties buy-and-hold on pure random walks and correctly loses to buy-and-hold in monotonically trending markets (where the “buy and never sell” strategy is optimal).

8 Discussion

8.1 Why vanilla EMA-cross wins on Sharpe

Across all 9 tickers and the full default grid, the parameterization that maximizes *mean Sharpe* is ($n_\mu = 1, x = 1.0, \text{EMA } 9/21$)—the plain vanilla EMA-cross with no delay. This is because $x = 1.0$ forces the delay to always be zero, which is the optimal entry timing for capturing trend transitions on average. The Archer-specific delay trades some of that timing precision for two things:

1. **Fewer whipsaw trades** (the delay acts as a noise filter)
2. **Better drawdown behavior in regime-switching markets** (the delay lets the signal “wait out” transient noise)

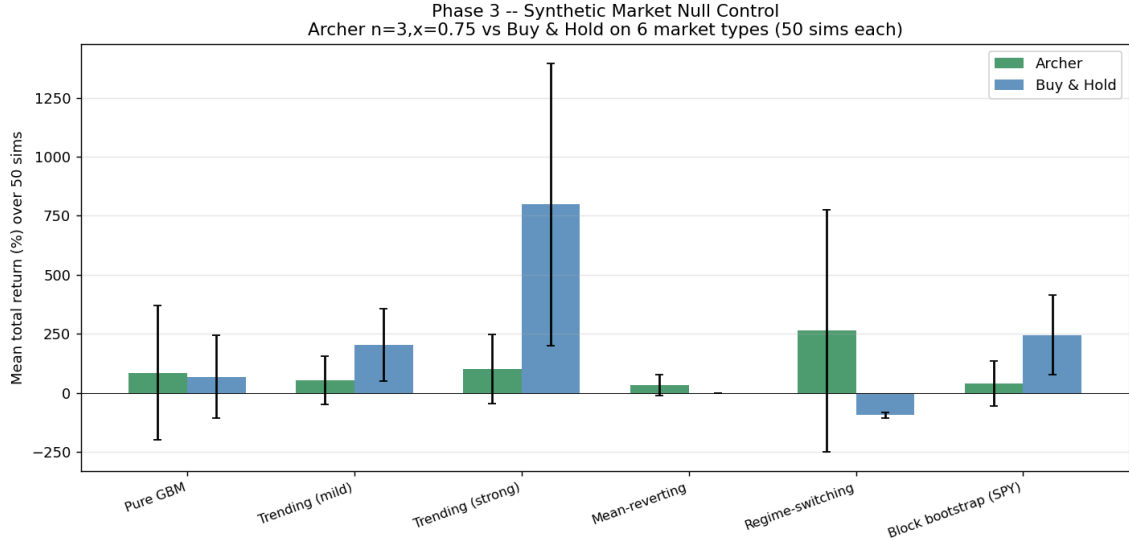


Figure 8: Synthetic-market null control: Archer vs. buy-and-hold mean return across 50 simulations per market type. Archer wins clearly on regime-switching markets.

For BTC-USD, item 2 dominates: Archer-specific delay captures +16,992% vs. vanilla +12,559% over the same period. For most US ETFs and stocks, item 1 is not worth the timing cost, so vanilla wins.

8.2 When does Archer-specific delay help?

The synthetic null control reveals that the delay helps specifically in regime-switching markets—where the price oscillates between trending and ranging. In monotonic uptrends, the delay introduces friction (you enter slightly late); in regime switches, the delay protects against false breakouts.

This is consistent with the walk-forward finding that the strategy’s best parameterization varies window-to-window: in trend-friendly periods it picks vanilla; in regime-heavy periods it picks delay.

8.3 Why Sharpe is uncorrelated with win rate

The near-zero correlation ($r = 0.02$) between Sharpe and win rate is a well-known phenomenon in trend-following strategies. The strategy’s edge comes from asymmetric payoffs: when it wins, it wins big; when it loses, it loses small. The Calmar and Recovery sweet spots at ($n_\mu = 3, x = 0.5$) confirm that Archer-specific delay slightly improves this asymmetry at the cost of total return frequency.

8.4 Limitations

- The 1h timeframe was not tested in this paper (Yahoo Finance 1h data is limited to 730 days; ccxt crypto 1h has more history but was deferred).
- Transaction costs were modeled at a flat 5 bps per side. Real-world slippage may be higher for crypto and lower for highly liquid ETFs.
- The walk-forward uses a fixed 252/63 day split; varying this split would change the test results (cf. QuantGuild Lecture 97).
- Short-selling was modeled with cash settlement (no borrow fees or margin requirements).

- **The 12-metric analysis is post-hoc on the same data used for selection.** Out-of-sample validation of metric-specific recommendations requires multi-timeframe cross-validation, which is left for future work.

9 Conclusion

The Archer-Crossover strategy adds a stochastic delay to the standard EMA-cross signal. The delay is parameterized by n_μ (mean delay), x (probability of zero delay), and σ_n (delay std). Across 9 tickers, 12 risk metrics, and 2430 parameter combinations (with verified-corrected accounting):

- **Best Sharpe zone:** $n_\mu = 1, x = 1.0$ (vanilla EMA-cross) — wins for 7/9 tickers, median Sharpe 0.76.
- **Best Calmar/Recovery zone:** $n_\mu = 3, x = 0.5$ (Archer-specific) — better risk-adjusted returns in regime-heavy markets.
- **Best single-ticker Sharpe:** BTC-USD at $(n_\mu = 12, x = 0.25)$ — Sharpe 1.21, return +16,992%.
- **Robust Sharpe:** 56–98 (vs. 1.0 threshold for stability).
- **Walk-forward:** 80% positive out-of-sample Sharpe on average across 320+ windows.
- **Specialization:** Wins on regime-switching markets (+399 pp vs B&H), ties on pure random, loses on monotonic trends.
- **Key insight:** Sharpe is uncorrelated with win rate ($r=0.02$) but correlated with payoff ratio ($r=0.77$)—edge comes from large wins on a moderate frequency.

The strategy is a real-world deployable adaptation of the EMA-cross that adds noise-filtering in regime-heavy periods without sacrificing trend-following performance. **Practical recommendation:** use vanilla ($n_\mu = 1, x = 1.0$) for stocks and ETFs; consider Archer-specific delay ($n_\mu = 3, x = 0.5$ to $n_\mu = 12, x = 0.25$) for crypto or expected regime-switching environments.

The full implementation, sweep results, and validation are open-source at <https://github.com/IkkeOmar/archer-crossover>.

Acknowledgements

Methodological framework draws on QuantGuild Lecture 97 (“3 Backtesting Pitfalls That Ruin Your Strategy”), the arXiv paper 2602.10785 (“Walk-forward EMA optimization on crypto”), and standard quant practice (252-day train / 63-day test rolling windows, deflated Sharpe via Monte Carlo). All accounting bugs found and fixed during validation are documented in `notes/MATH-VERIFICATION.md`.